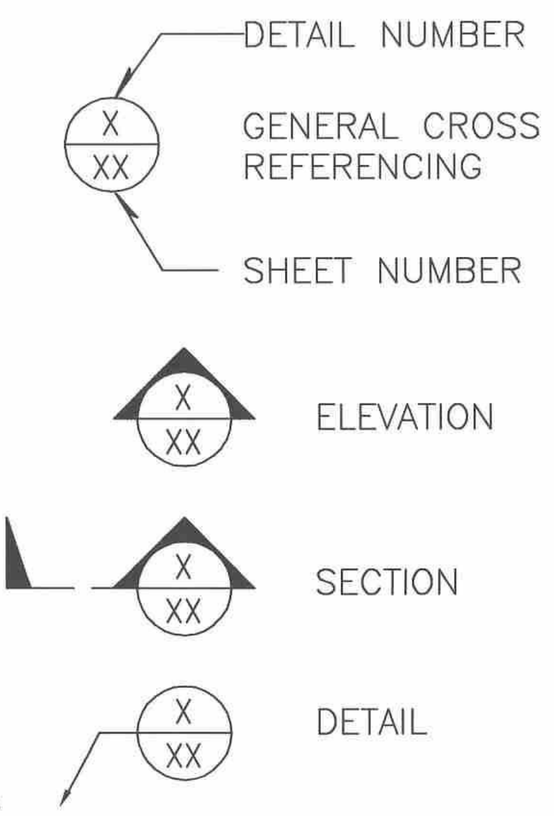
## ABBREVIATIONS & SYMBOLS

| | |
|---|---|
| AC | ASPHALTIC CONCRETE |
| ADJ. | ADJUSTABLE |
| AFF | ABOVE FINISHED FLOOR |
| AFG | ABOVE FINISHED GRADE |
| ALUM. | ALUMINUM |
| AMSL | ABOVE MEAN SEA LEVEL |
| APPROX. | APPROXIMATE |
| AWG | AMERICAN WIRE GAUGE |
| BCW | BARE COPPER WIRE |
| BLKG. | BLOCKING |
| CAB. | CABINET |
| CLG. | CEILING |
| CONC. | CONCRETE |
| CONT. | CONTINUOUS |
| Cu | COPPER |
| DIA. | DIAMETER |
| DWG. | DRAWING |
| E | EXISTING TO REMAIN |
| EA. | EACH |
| EL | ELEVATION |
| ELEC. | ELECTRICAL |
| EQ. | EQUAL |
| EQUIP. | EQUIPMENT |
| EXIST. | EXISTING |
| EXT. | EXTERIOR |
| F.F. | FINISHED FLOOR |
| GA. | GAUGE |
| GALV. | GALVANIZED |
| GND | GROUND |
| GPS | GEOSTATIONARY POSITIONING SYSTEM |
| INT. | INTERIOR |
| MAX. | MAXIMUM |
| MECH. | MECHANICAL |
| MFR. | MANUFACTURER |
| MIN. | MINIMUM |
| N.T.S. | NOT TO SCALE |
| O.C. | ON CENTER |
| OPP. | OPPOSITE |
| PL | PLATE |
| S.F. | SQUARE FOOT |
| SHT | SHEET |
| STL. | STEEL |
| SUSP. | SUSPENDED |
| TNND | TINNED |
| T.O.C. | TOP OF CONCRETE |
| TYP. | TYPICAL |
| U.O.N. | UNLESS OTHERWISE NOTED |
| WP | WEATHERPROOF |
| W.W.F. | WELDED WIRE FABRIC |
| W/ | WITH |
| & | AND |
| @ | AT |



## RF DATA SHEET

RF ENGINEER NAME: ALAN BUNGO
PHONE #: (303) 525-1729

DATE: 5/15/01

COMMENTS: ORDER MECHANICAL DOWNTILT BRACKETS FOR ALL ANTENNAS.

SITE NAME: CEMEX CEMENT PLANT
SITE NUMBER: BLD-021A

HAS R.F. DRIVE TESTING BEEN COMPLETED: YES: X NO:

LATTITUDE: 40°12'12.7" N
LONGITUDE: 105°14'7.9" W

STRUCTURE TYPE – PROPOSED (P) OR EXISTING (E)

| | |
|---|---|
| SELF SUPPORT: | SCREEN: |
| MONOPOLE: | OTHER (DESCRIBE BELOW): |
| LATTICE TOWER: | PENTHOUSE PARAPET |
| ROOFTOP: X | LUCENT BTS X |
| PICO: | |

IS THE SITE A SECOND CARRIER AT TIME OF CONSTRUCTION?
☐ YES ☒ NO

STRUCTURE LOADING (INCLUDING CO-LOCATORS, EXISTING AND PROPOSED EQUIPMENT

CELLULAR ANTENNAS

| AZIMUTH (DEGREES) | # PER SECTOR | ANTENNA CENTERLINE HEIGHT (FEET) | | | MANUFACTURER NAME/MODEL NUMBER | DOWN TILT (MECHANICAL) (DEGREE) | SEPARATION | | EXISTING OR PROPOSED (E OR P) |
|---|---|---|---|---|---|---|---|---|---|
| | | Tx | Rx | FUT. | | | VERTICAL (FEET) | HORIZONTAL (FEET) | |
| 80 | 2+1 | 130 | 130 | | FV65-17-00DP | 0 | | 10 | P |
| 200 | 2+1 | 130 | 130 | | FV65-17-00DP | 0 | | 10 | P |
| 320 | 2+1 | 130 | 130 | | FV65-17-00DP | 0 | | 10 | P |

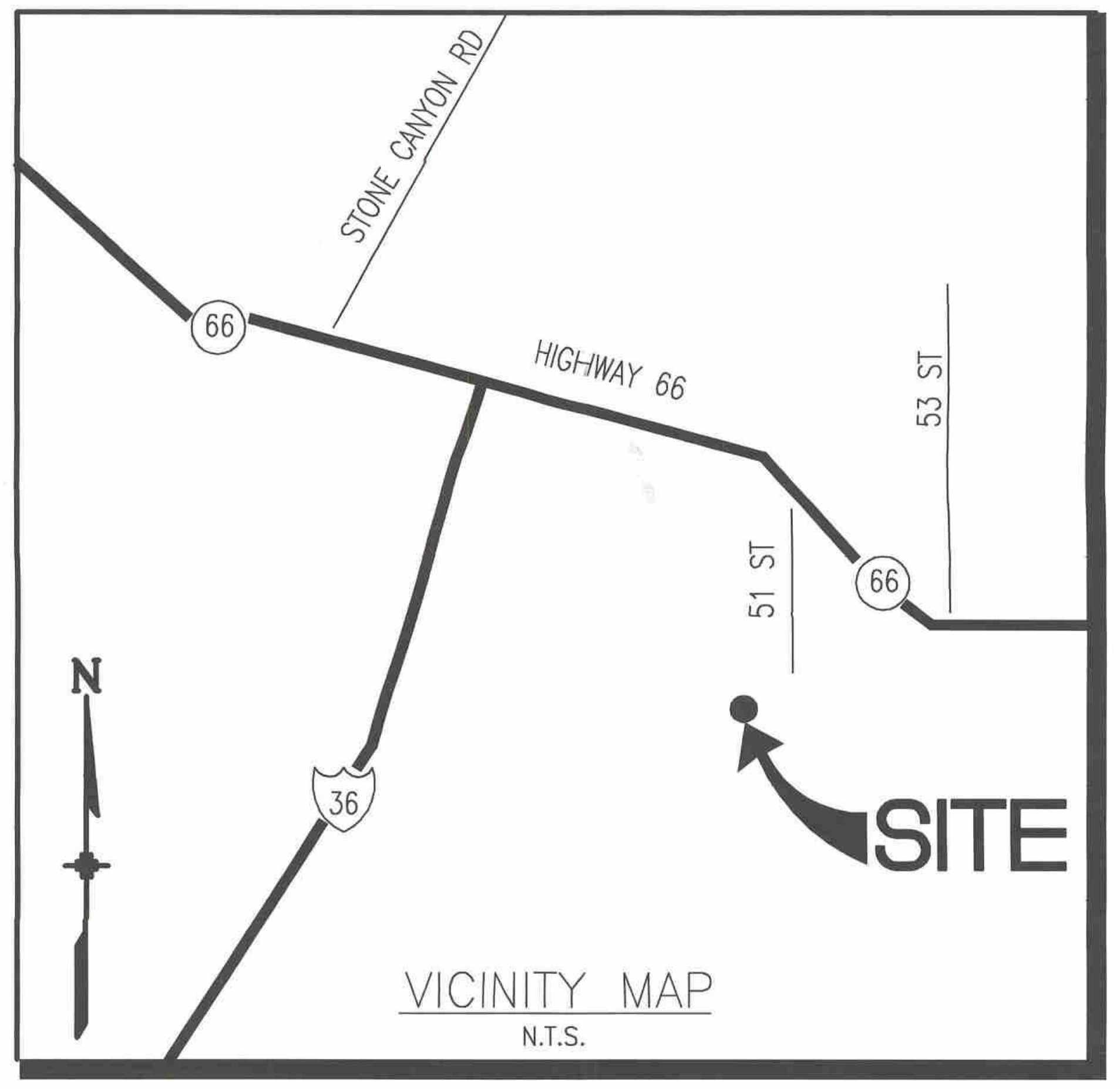
GPS ANTENNA

DESCRIBE LOACTION AND ELEVATION: MOUNTED ON B.T.S. EQUIPMENT ON ROOF

MICROWAVE ANTENNAS

| AZIMUTH (DEGREES) | DIAMETER (FEET) | ANTENNA CENTERLINE HEIGHT (FEET) | MANUFACTURER NAME/MODEL NUMBER | DOWN TILT (MECHANICAL) (DEGREE) | SEPARATION | | EXISTING OR PROPOSED (E OR P) |
|---|---|---|---|---|---|---|---|
| | | | | | VERTICAL (FEET) | HORIZONTAL (FEET) | |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |

RF SIGNAGE:
POSTING REQUIRED @ ACCESS ☐ YES ☒ NO
POSTING REQUIRED ON ANTENNA FACE ☐ YES ☒ NO



VICINITY MAP
N.T.S.

## GENERAL NOTES

1. ADA COMPLIANCE: PCS FACILITY IS NOT STAFFED AND NOT NORMALLY OCCUPIED.

## LEGAL DESCRIPTION

SITE PARCEL NO: 1203-28-0-00-001

SITE NAME: CEMEX CEMENT PLANT

LEGAL DESCRIPTION: SECTION TOWNSHIP RANGE: 21-3N-70, 28-3N-70, 29-3N-70
SE 1/4 SE 1/4 20-3N-70 & PT W 1/2 27-3N-70 & E 500 FT OF LESS.19 ACS M/L TO BOCO & LESS PTS TO US LESS PT TO HWY DEPT TOTAL 757.85 ACS SPLITS 128969 & 142930 TO HWY DEPT

## PROJECT SUMMARY

- INSTALLATION OF B.T.S EQUIPMENT CABINETS ON EXISTING CEMENT PLANT ROOFTOP
- INSTALLATION OF 3 ANTENNA ARRAYS WITH 2 ANTENNA EACH WITH 1 FUTURE ON FACADE OF PLANT
- POWER AND TELCO FROM EXISTING SERVICES.
- GPS MOUNTED ON BTS EQUIPMENT ON ROOF

## PROJECT INFORMATION

SITE ADDRESS: 5134 UTE HIGHWAY
LYONS, COLORADO 80540-0529

ZONING CLASSIFICATION: GI (GENERAL INDUSTRIAL)

AREA OF CONSTRUCTION (SQ FT): 204 SF

OCCUPANCY CLASSIFICATION: U (UNMANNED TELECOMMUNICATIONS FACILITY)

## PROJECT DIRECTORY

**OWNER:** SOUTHDOWN INC
5134 UTE HIGHWAY
LYONS, COLORADO 80540-0529

CONTACT: JOHN LOHR,PLANT MGR. (303) 823-2100

**APPLICANT:** QWEST WIRELESS LLC
4301 E. COLFAX AVE., ROOM 314
DENVER, CO 80220

CONTACT: DALLAS KILAND (303) 398-8500

**AGENT:** QWEST WIRELESS LLC
4301 E. COLFAX AVE., ROOM 314
DENVER, CO 80220

CONTACT: KARYN KARLSON (303) 587-6146

**ENGINEER:** QWEST WIRELESS LLC
4301 E. COLFAX AVE., ROOM 314
DENVER, CO 80220
(303) 398-8497 FAX

CONTACT: RANDALL STEINKE (720) 260-7093

## DRAWINGS

| | |
|---|---|
| T-1 | TITLE SHEET |
| A-1 | SITE PLAN AND ENLARGED SITE PLAN |
| A-2 | ELEVATION |
| A-3 | DETAILS |
| E-1 | ONE-LINE DIAGRAM |
| E-2 | ELECTRICAL PLAN |
| E-3 | DETAILS |

| NO. | DATE | REVISION | BY | CHKD | APVLS |
|---|---|---|---|---|---|
| △A | 10 12 01 | ISSUED FOR QWEST REVIEW | IR | RLS | |
| △0 | 10 23 01 | ISSUED FOR CONSTRUCTION/PERMIT | IR | RLS | |
| △ | | | | | |
| △ | | | | | |
| △ | | | | | |
| △ | | | | | |
| △ | | | | | |

STATE OF COLORADO
RANDALL L. STEINKE
C-3371
LICENSED ARCHITECT

11.19.01

Qwest® Wireless™

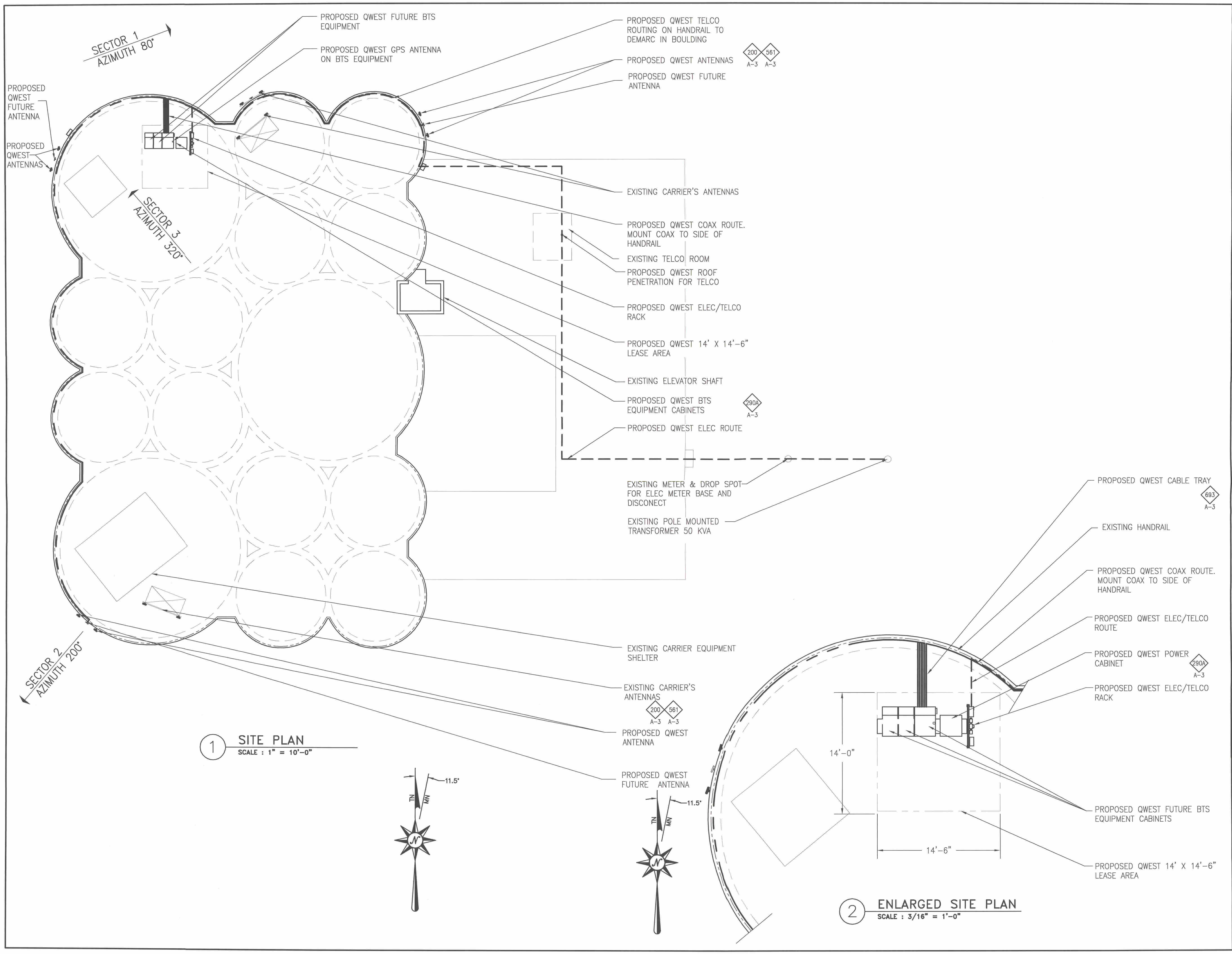
SITE INFORMATION
SOUTHDOWN CEMENT PLANT
5134 UTE HIGHWAY
LYONS, COLORADO 80540-0529
BLD-021A

DESIGN TYPE
ROOFTOP EQUIPMENT
AND
ANTENNAS

SHEET TITLE
TITLE SHEET

SHEET NUMBER: T-1
REV.: 0

DWG NAME: T-1
DATE: 10-23-01


SECTOR 1
AZIMUTH 80°
PROPOSED QWEST FUTURE ANTENNA
PROPOSED QWEST ANTENNAS
SECTOR 3
AZIMUTH 320°
SECTOR 2
AZIMUTH 200°
PROPOSED QWEST FUTURE BTS EQUIPMENT
PROPOSED QWEST GPS ANTENNA ON BTS EQUIPMENT
PROPOSED QWEST TELCO ROUTING ON HANDRAIL TO DEMARC IN BOULDING
PROPOSED QWEST ANTENNAS
200 A-3
561 A-3
PROPOSED QWEST FUTURE ANTENNA
EXISTING CARRIER'S ANTENNAS
PROPOSED QWEST COAX ROUTE. MOUNT COAX TO SIDE OF HANDRAIL
EXISTING TELCO ROOM
PROPOSED QWEST ROOF PENETRATION FOR TELCO
PROPOSED QWEST ELEC/TELCO RACK
PROPOSED QWEST 14' X 14'-6" LEASE AREA
EXISTING ELEVATOR SHAFT
PROPOSED QWEST BTS EQUIPMENT CABINETS
290A A-3
PROPOSED QWEST ELEC ROUTE
EXISTING METER & DROP SPOT FOR ELEC METER BASE AND DISCONECT
EXISTING POLE MOUNTED TRANSFORMER 50 KVA
EXISTING CARRIER EQUIPMENT SHELTER
EXISTING CARRIER'S ANTENNAS
200 A-3
561 A-3
PROPOSED QWEST ANTENNA
PROPOSED QWEST FUTURE ANTENNA
TN
MN
11.5°
N
TN
MN
11.5°
N
PROPOSED QWEST CABLE TRAY
693 A-3
EXISTING HANDRAIL
PROPOSED QWEST COAX ROUTE. MOUNT COAX TO SIDE OF HANDRAIL
PROPOSED QWEST ELEC/TELCO ROUTE
PROPOSED QWEST POWER CABINET
290A A-3
PROPOSED QWEST ELEC/TELCO RACK
14'-0"
14'-6"
PROPOSED QWEST FUTURE BTS EQUIPMENT CABINETS
PROPOSED QWEST 14' X 14'-6" LEASE AREA

① SITE PLAN
SCALE : 1" = 10'-0"

② ENLARGED SITE PLAN
SCALE : 3/16" = 1'-0"

| NO. | DATE | REVISION | BY | CHKD | APVLS |
|---|---|---|---|---|---|
| A | 10<br>12<br>01 | ISSUED FOR QWEST REVIEW | IR | RLS | |
| 0 | 10<br>23<br>01 | ISSUED FOR CONSTRUCTION/PERMIT | IR | RLS | |
| | | | | | |
| | | | | | |
| | | | | | |
| | | | | | |
| | | | | | |

RANDALL L. STEINKE

11.19.01

Qwest® Wireless™

SITE INFORMATION

SOUTHDOWN CEMENT PLANT
5134 UTE HIGHWAY
LYONS, COLORADO 80540-0529
BLD-021A

DESIGN TYPE

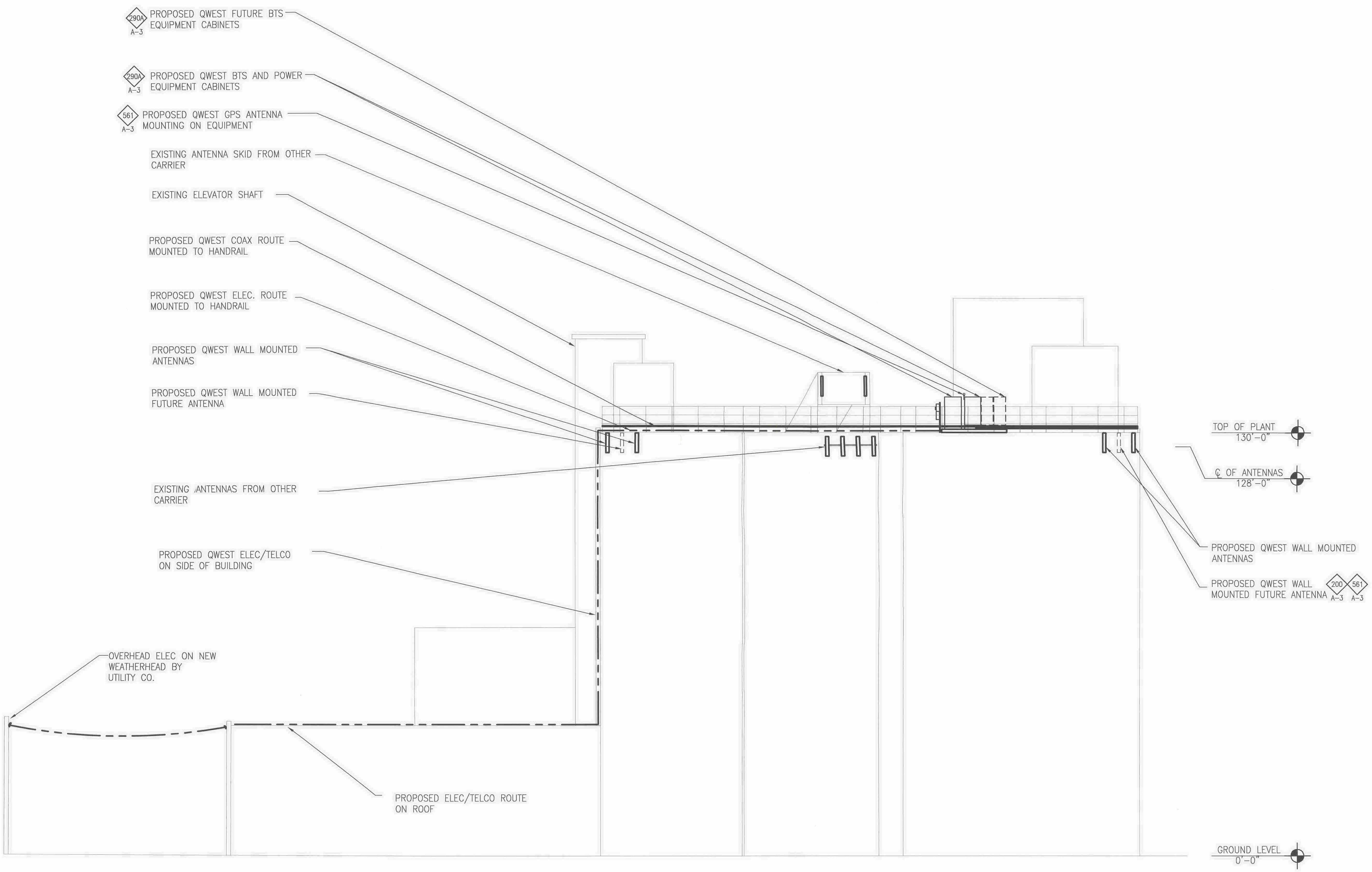
ROOFTOP EQUIPMENT
AND
ANTENNAS

SHEET TITLE

SITE PLAN
AND
ENLARGED SITE PLAN

| SHEET NUMBER | REV. |
|---|---|
| A-1 | 0 |

DWG NAME: A-1
DATE: 10-23-01


290A A-3 PROPOSED QWEST FUTURE BTS EQUIPMENT CABINETS
290A A-3 PROPOSED QWEST BTS AND POWER EQUIPMENT CABINETS
561 A-3 PROPOSED QWEST GPS ANTENNA MOUNTING ON EQUIPMENT
EXISTING ANTENNA SKID FROM OTHER CARRIER
EXISTING ELEVATOR SHAFT
PROPOSED QWEST COAX ROUTE MOUNTED TO HANDRAIL
PROPOSED QWEST ELEC. ROUTE MOUNTED TO HANDRAIL
PROPOSED QWEST WALL MOUNTED ANTENNAS
PROPOSED QWEST WALL MOUNTED FUTURE ANTENNA
EXISTING ANTENNAS FROM OTHER CARRIER
PROPOSED QWEST ELEC/TELCO ON SIDE OF BUILDING
OVERHEAD ELEC ON NEW WEATHERHEAD BY UTILITY CO.
PROPOSED ELEC/TELCO ROUTE ON ROOF
TOP OF PLANT 130'-0"
℄ OF ANTENNAS 128'-0"
PROPOSED QWEST WALL MOUNTED ANTENNAS
PROPOSED QWEST WALL MOUNTED FUTURE ANTENNA 200 A-3 561 A-3
GROUND LEVEL 0'-0"

① NORTH ELEVATION

SCALE : 1/8" = 1'-0"

| NO. | DATE | REVISION | BY | CHKD | APVLS |
|---|---|---|---|---|---|
| A | 10<br>12<br>01 | ISSUED FOR QWEST REVIEW | IR | RLS | |
| 0 | 10<br>23<br>01 | ISSUED FOR CONSTRUCTION/PERMIT | IR | RLS | |
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| | | | | | |
| | | | | | |

11.14.01

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SITE INFORMATION

SOUTHDOWN CEMENT PLANT
5134 UTE HIGHWAY
LYONS, COLORADO 80540–0529
BLD–021A

DESIGN TYPE

ROOFTOP EQUIPMENT
AND
ANTENNAS

SHEET TITLE

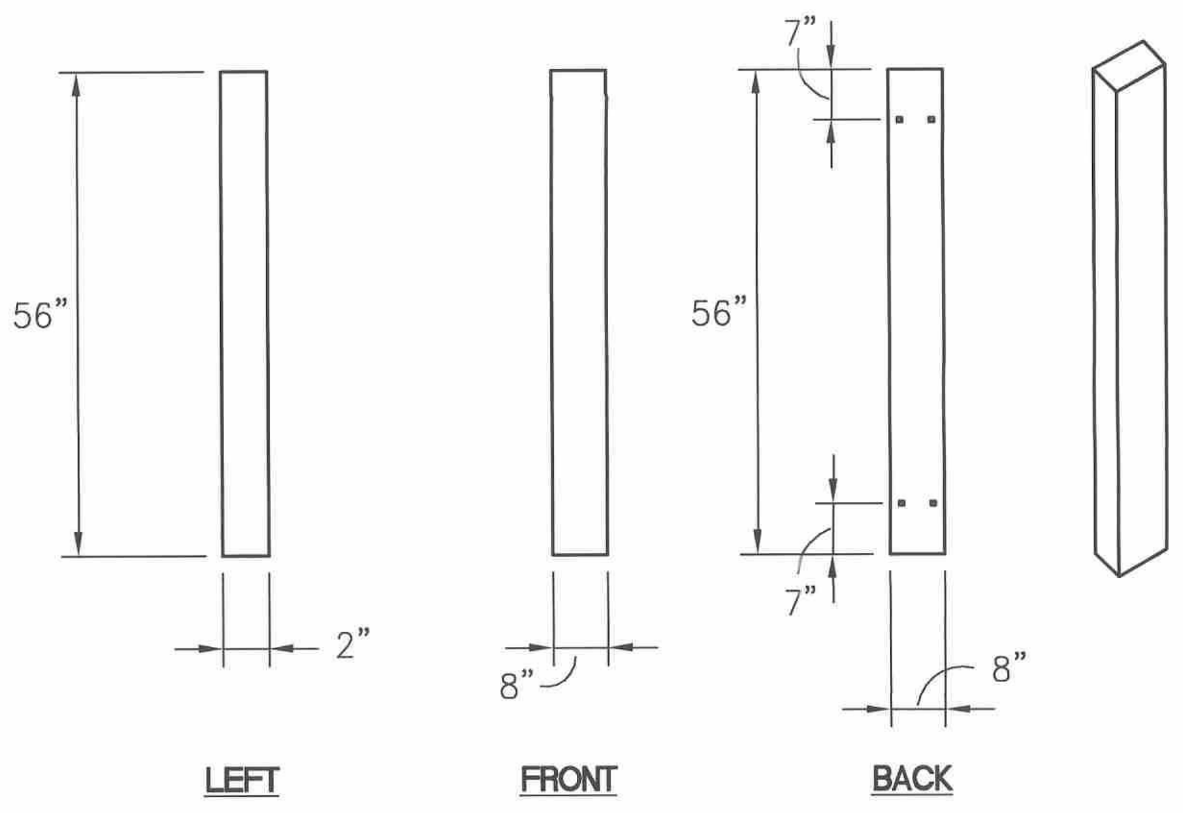
ELEVATION

SHEET NUMBER

A–2 0

DWG NAME: A–2

DATE: 10–23–01

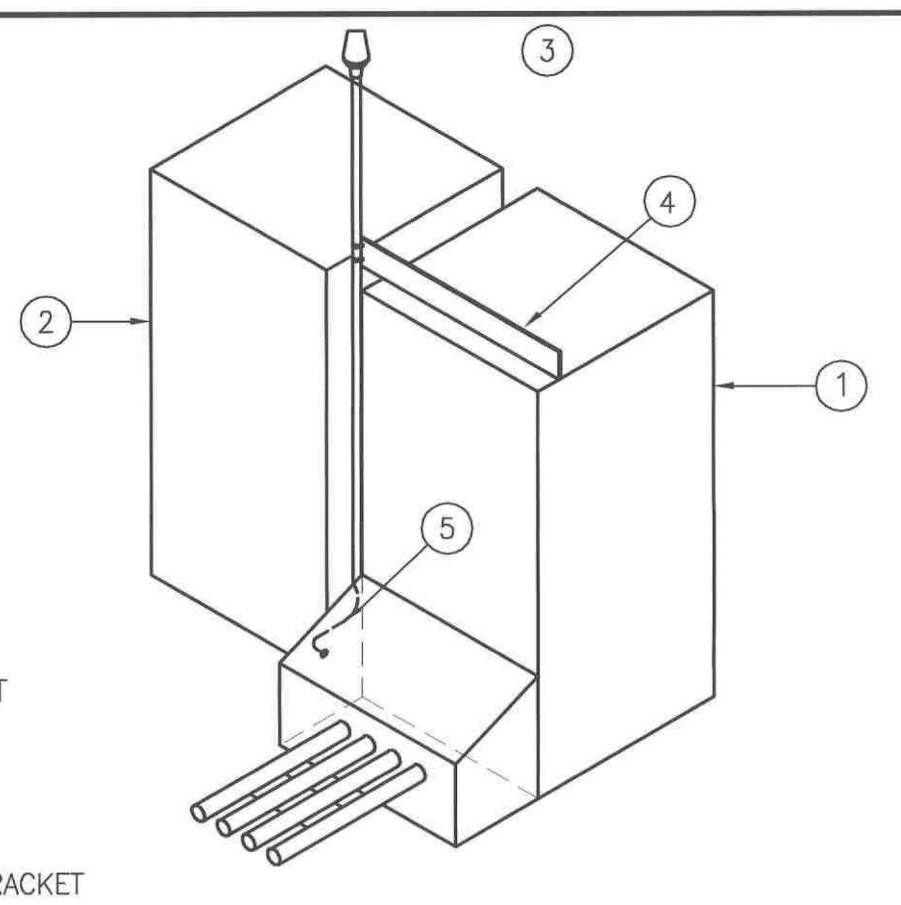


56"
7"
56"
7"
2"
8"
8"
LEFT
FRONT
BACK

NOTE: ANTENNA IS FV-65-17-00DP

## 200 TYPICAL ANTENNA

SCALE: N.T.S.

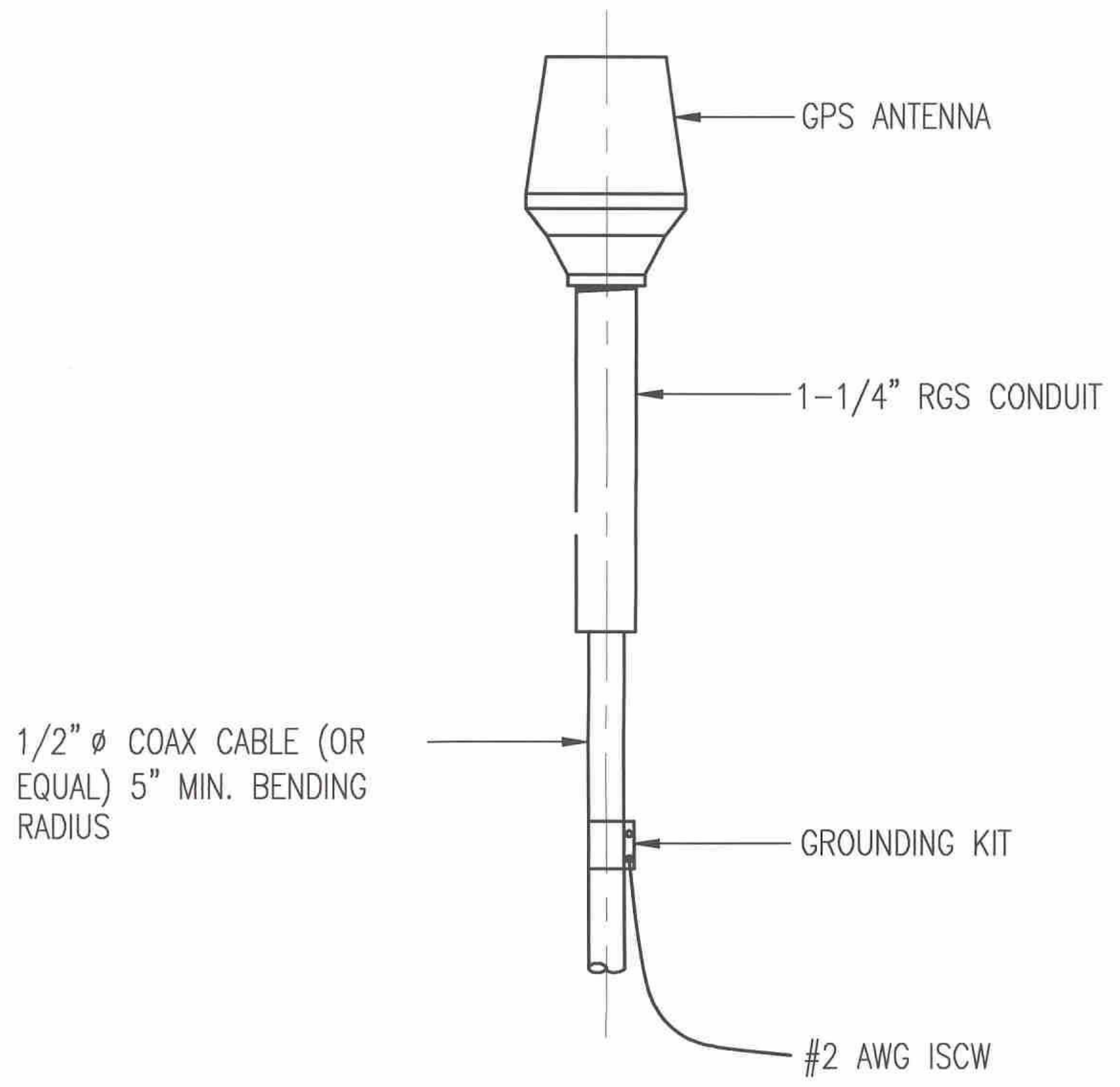


**KEY NOTES:**

① MINI-CELL CABINET
② POWER CABINET
③ GPS ANTENNA
④ GPS MOUNTING BRACKET
⑤ GPS COAX CABLE

## 290A CABINET MOUNTED GPS ANTENNA

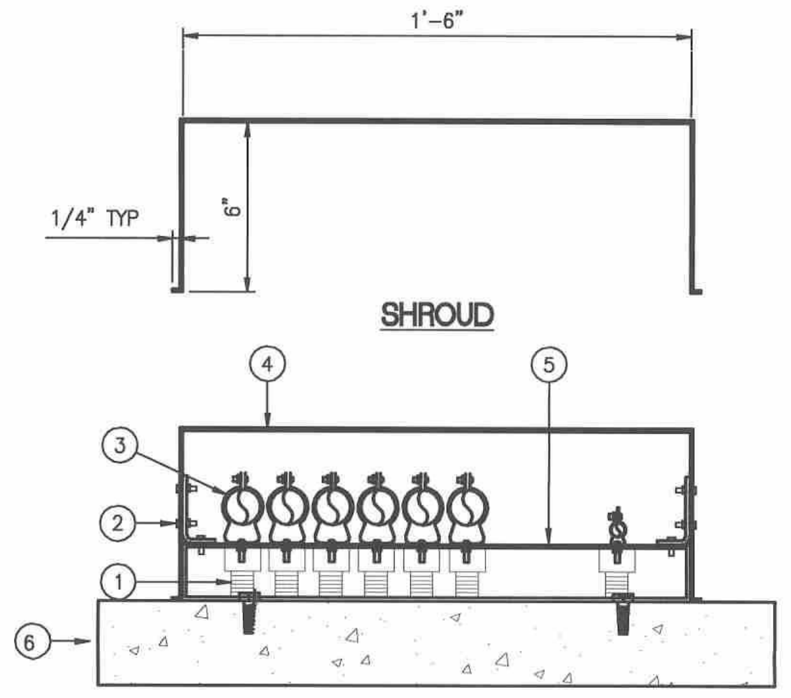
SCALE: N.T.S.



GPS ANTENNA
1-1/4" RGS CONDUIT
1/2" ø COAX CABLE (OR EQUAL) 5" MIN. BENDING RADIUS
GROUNDING KIT
#2 AWG ISCW

## 561 GPS ANTENNA

SCALE: N.T.S.



1'-6"
1/4" TYP
6"
SHROUD
①
②
③
④
⑤
⑥

**KEY NOTES:**

① UNISTRUT 3/8 " DIA P1008 SPRING-LOADED NUT (TYP PER HANGER)
② ATTACH SHROUD TO UNISTRUT P1538A (4'-0" OC W/ (2) #10 TEK'S, TYP EACH SIDE)
③ COAX (SIZE & QUANTITY AS REQUIRED) W/ NON-INSULATING HANGERS @ 4'-0" OC (TYP)
④ 20 GA GALV METAL SHROUD (IF REQUIRED, PAINT W/ 2 COATS OF ALKYD EXTERIOR PAINT; FINISH PER REQUEST)
⑤ UNISTRUT P1000x18" LONG - ATTACH TO CONC PAVERS REQUIRED FOR ROOFTOP CONDITION
⑥ 4"x4"x16" CONCRETE PAVERS

## 693A CABLE SUPPORT ON CONC PAVERS W/SHROUD

SCALE: N.T.S.

| NO. | DATE | REVISION | BY | CHKD | APVLS |
|---|---|---|---|---|---|
| A | 10<br>12<br>01 | ISSUED FOR QWEST REVIEW | IR | RLS | |
| 0 | 10<br>23<br>01 | ISSUED FOR CONSTRUCTION/PERMIT | IR | RLS | |
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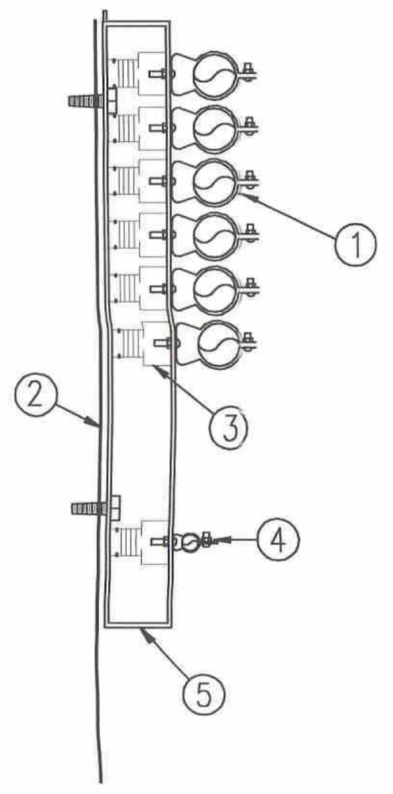
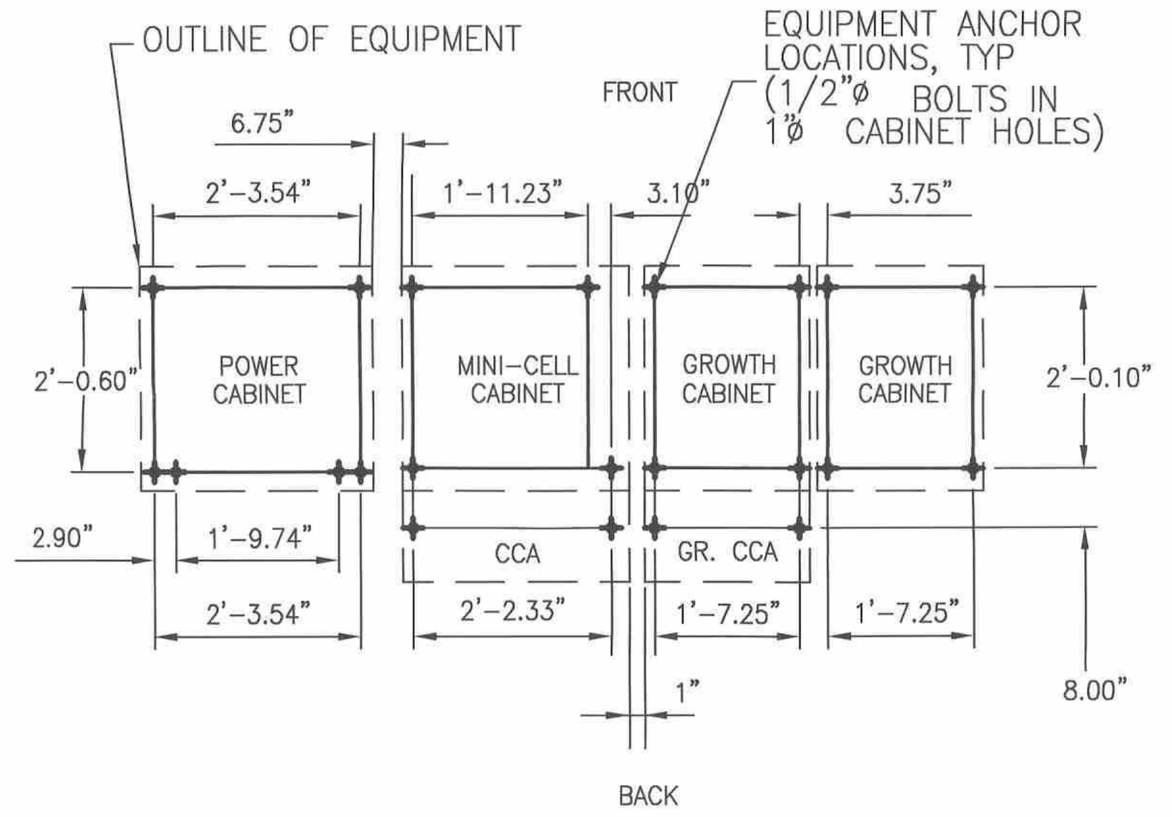


**KEY NOTES:**

① COAX (SIZE AND QUANTITY AS REQUIRED) W/ NON- INSULATING HANGERS @ 4'-0" O.C. (TYPICAL).
② GUARDRAIL
③ UNISTRUT 3/8" DIA. P1008 SPRING-LOADED NUT (TYP PER HANGER)
④ 1/2" GPS COAX W/ NON-INSULATED HANGER @ 48" O.C . (TYPICAL)
⑤ UNISTRUT P1000x18" LONG ATTACH TO PAVERS W/ (2)-3/8 " DIA. EXPANSION BOLTS.

## 557 CABLE SUPPORT ON GUARDRAIL

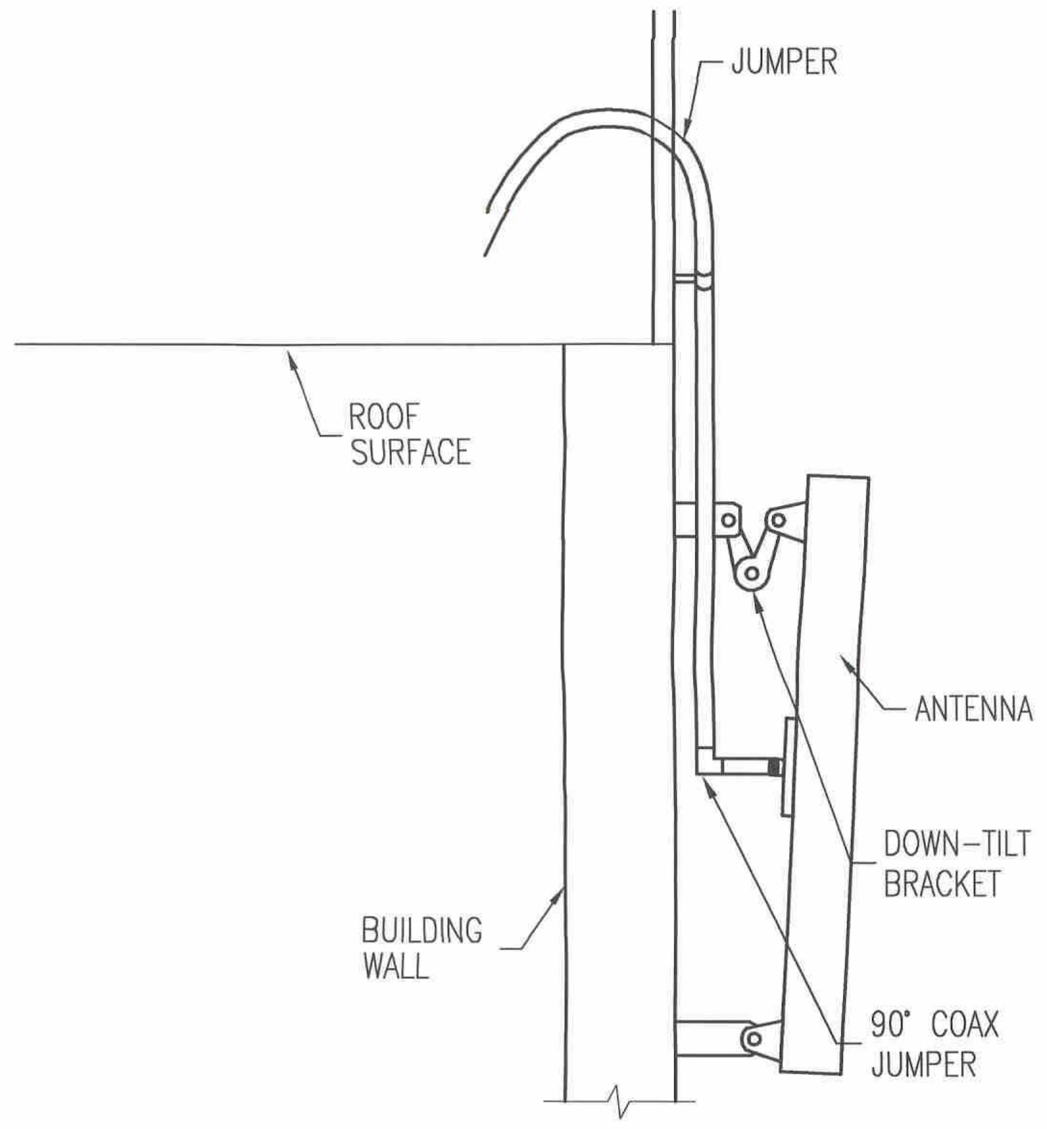
SCALE: N.T.S.



OUTLINE OF EQUIPMENT
EQUIPMENT ANCHOR LOCATIONS, TYP (1/2"ø BOLTS IN 1"ø CABINET HOLES)
FRONT
6.75"
2'-3.54"
1'-11.23"
3.10"
3.75"
2'-0.60"
POWER CABINET
MINI-CELL CABINET
GROWTH CABINET
GROWTH CABINET
2'-0.10"
2.90"
1'-9.74"
CCA
GR. CCA
2'-3.54"
2'-2.33"
1'-7.25"
1'-7.25"
1"
8.00"
BACK

## 910 EQUIPMENT ANCHOR PATTERN

SCALE: N.T.S.



JUMPER
ROOF SURFACE
ANTENNA
DOWN-TILT BRACKET
BUILDING WALL
90° COAX JUMPER

## 260 BUILDING MOUNT ANTENNA(CONCRETE OR MASONRY WALL)

SCALE: N.T.S.

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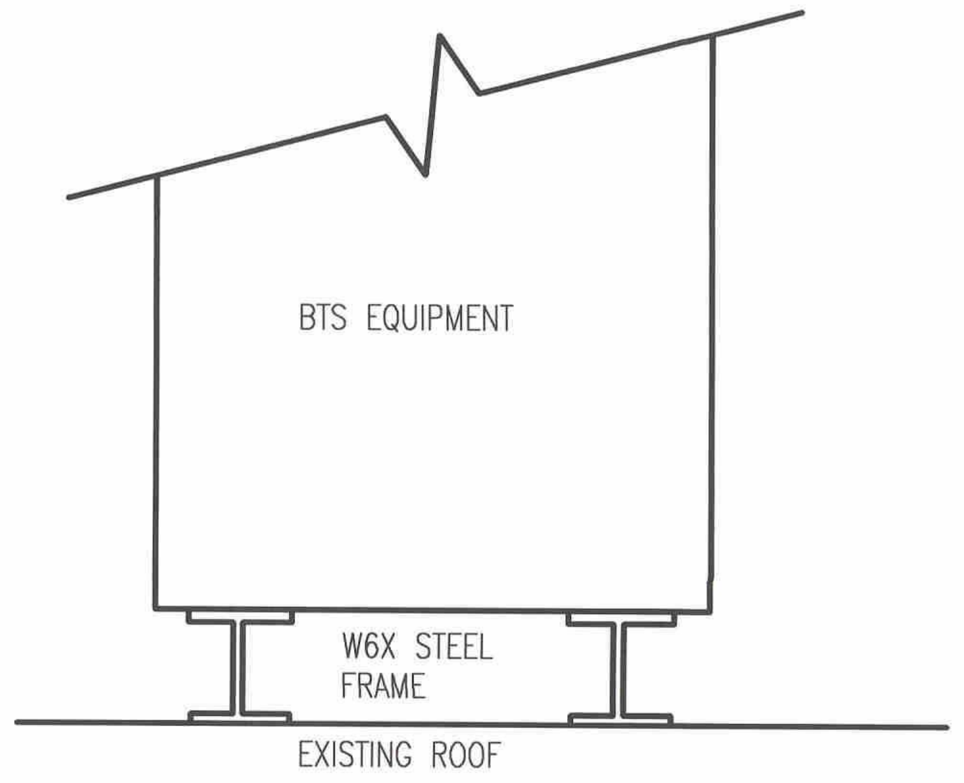


BTS EQUIPMENT
W6X STEEL FRAME
EXISTING ROOF

## 920 BTS EQUIPMENT SUPPORT STRUCTURE

SCALE: N.T.S.

SITE INFORMATION

SOUTHDOWN CEMENT PLANT
5134 UTE HIGHWAY
LYONS, COLORADO 80540-0529
BLD-021A

DESIGN TYPE

ROOFTOP EQUIPMENT
AND
ANTENNAS

SHEET TITLE

DETAILS

SHEET NUMBER: A-3

REV.: 0

DWG NAME: A3

DATE: 10-23-01

## ELECTRICAL SYMBOLS

| | |
|---|---|
| ⊗ | GROUND ROD |
| ■ | CADWELD TYPE CONNECTION |
| ● | COMPRESSION TYPE CONNECTION |
| -·—·—·—·- | GROUND WIRE |

## ABBREVIATIONS

| | |
|---|---|
| AWG | AMERICAN WIRE GAUGE |
| BCW | BARE COPPER WIRE (SOLID TINNED) |
| GPS | GEOSTATIONARY POSITIONING SYSTEM |
| PCS | PERSONAL COMMUNICATION SERVICE |
| PPC | POWER PROTECTION CABINET |
| PRC | PRIMARY RADIO CABINET |
| RWY | RACEWAY |
| TYP. | TYPICAL |
| RGS | RIGID GALVANIZED STEEL |
| EMT | ELECTRICAL METALLIC TUBING |
| DWG | DRAWING |
| IGR | INTERIOR GROUND RING (HALO) |
| ACCA | ANTENNA CABLE COVER ASSEMBLY |
| BTS | BASE TRANSMISSION SYSTEM |
| GEN | GENERATOR |
| GR | GROWTH |
| BSCW | BARE STRANDED COPPER WIRE |
| ISCW | INSULATED STRANDED COPPER WIRE |

**ELECTRICAL SYMBOLS AND ABBREVIATIONS** SCALE : N.T.S.

## NOTES:

1. DO NOT INSTALL ANTENNA GROUND KIT ON CABLE BEND.

| FROM/TO | AZIMUTH 0° | DOWNTILT=NA | | AZIMUTH 120° | DOWNTILT=NA | | AZIMUTH 240° | DOWNTILT=NA | |
|---|---|---|---|---|---|---|---|---|---|
| | JUMPER & CABLE PART NUMBER | LENGTH (FT)* | | JUMPER & CABLE PART NUMBER | LENGTH (FT)* | | JUMPER & CABLE PART NUMBER | LENGTH (FT)* | |
| | | RX/TX | RX | | RX/TX | RX | | RX/TX | RX |
| JUMPER CABLE TO ANTENNA CABLE | L4-P10-PDMDM-3 | 3' | 3' | L4-P10-PDMDM-3 | 3' | 3' | L4-P10-PDMDM-3 | 3' | 3' |
| CABLE TO TERMINATION POINT | LDF5-50A | 55' | 55' | LDF5-50A | 90' | 90' | LDF7-50A | 180' | 180' |
| TERMINATION POINT TO CABLE COVER ASSEMBLY | FSJ-50A | 3' | 3' | FSJ-50A | 3' | 3' | FSJ-50A | 3' | 3' |

* ESTIMATED LENGTH FROM AVAILABLE INFORMATION. FIELD ENGINEER TO UPDATE ON AS BUILT DRAWINGS.

### 670 ANTENNA COAXIAL CABLE SCHEDULE

SCALE : N.T.S.

## QWEST CABINET PANEL SCHEDULE

100A MAIN PANEL INSIDE POWER CABINET 10,000 A.I.C.
120/240 VOLTS, 1 PHASE 3 WIRES 100 AMP. BUSS

| LOAD | WATTS | P | CB | CKT | PH | CKT | CB | P | WATTS | LOAD |
|---|---|---|---|---|---|---|---|---|---|---|
| GFI RECEPT. | 180 | 1 | 20 | 1 | A | 2 | 60 | 2 | 4888 | PWR. CAB. |
| SPARE | | 1 | 20 | 3 | B | 4 | 60 | 2 | 4888 | PWR. CAB. |
| SPARE | | 1 | 20 | 5 | A | 6 | 20 | 1 | | SPARE |
| SPACE | | 1 | | 7 | B | 8 | 20 | 1 | | SPARE |
| SPACE | | 1 | | 9 | A | 10 | | | | SPACE |
| SPACE | | 1 | | 11 | B | 12 | | | | SPACE |

TOTAL DEMAND 9,956VA, 41 AMPERE

## FAULT CURRENT CALCULATIONS

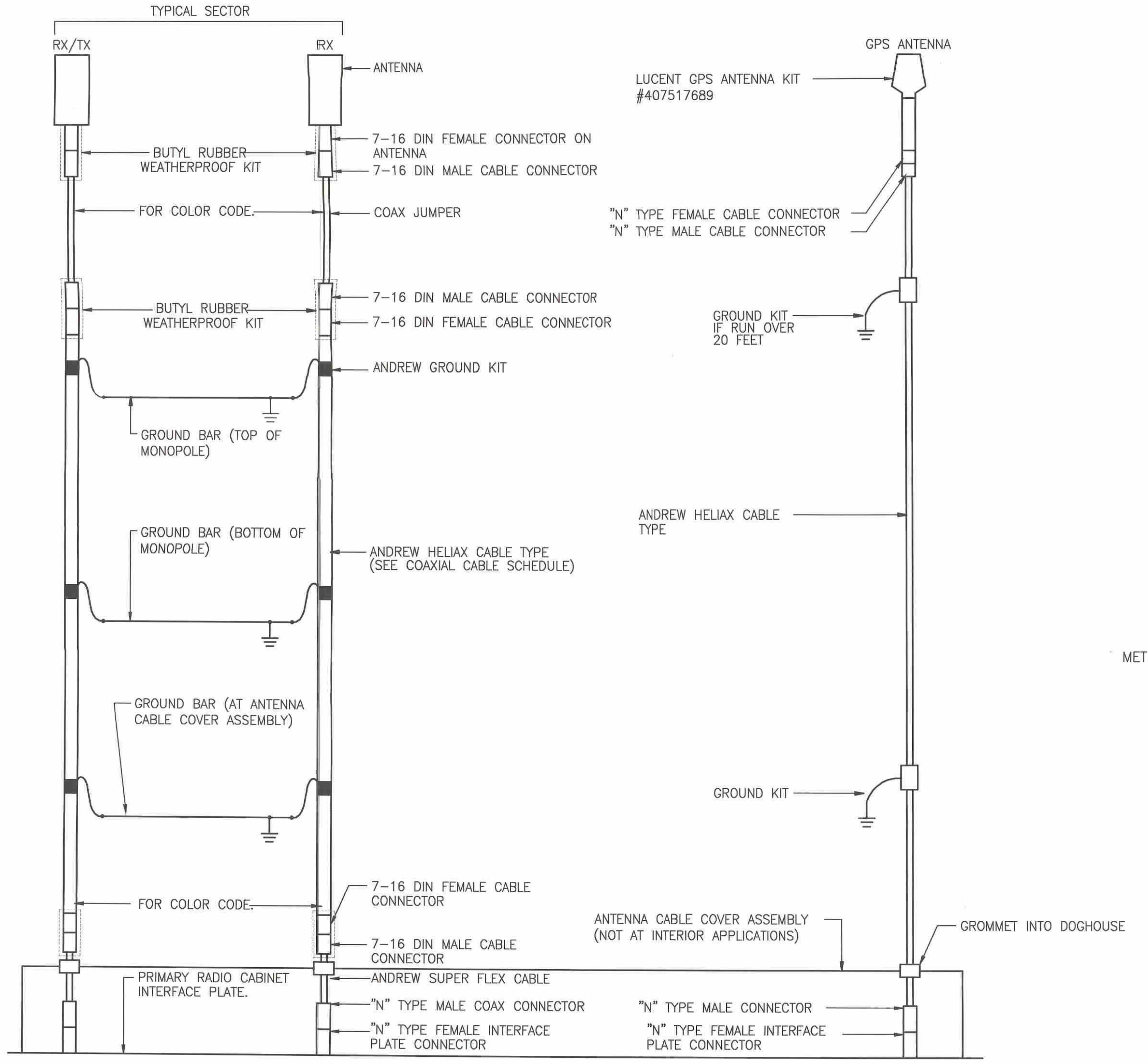
MAXIMUM AVAILABLE FAULT CURRENT AT THE SECONDARY OF EXISTING 50 KVA, 120/240V, 1PH, 3W POWER CO POLE MOUNTED TRANSFORMER IS 14,800 AMPS.
LENGTH OF RUN OF #4/0 CU FROM TRANSFORMER TO A NEW POWER CABINET IS 350 FT.
L=350 FT., C= 15,082

$$f = \frac{2 \text{ X } L \text{ X } IS.C.}{C \text{ X } E} = \frac{2 \text{ X } 350 \text{ X } 14,800}{15,082 \text{ X } 240} = 2.86$$

$$M = \frac{1}{1 + f} = \frac{1}{1 + 2.86} = 0.26$$

IS.C. AT POWER CABINET = IS.C. X M = 14,800 X 0.26
= 3,848 AMPS

NEW POWER CABINET SHALL BE RATED A MINIMUM 10,000 A.I.C.

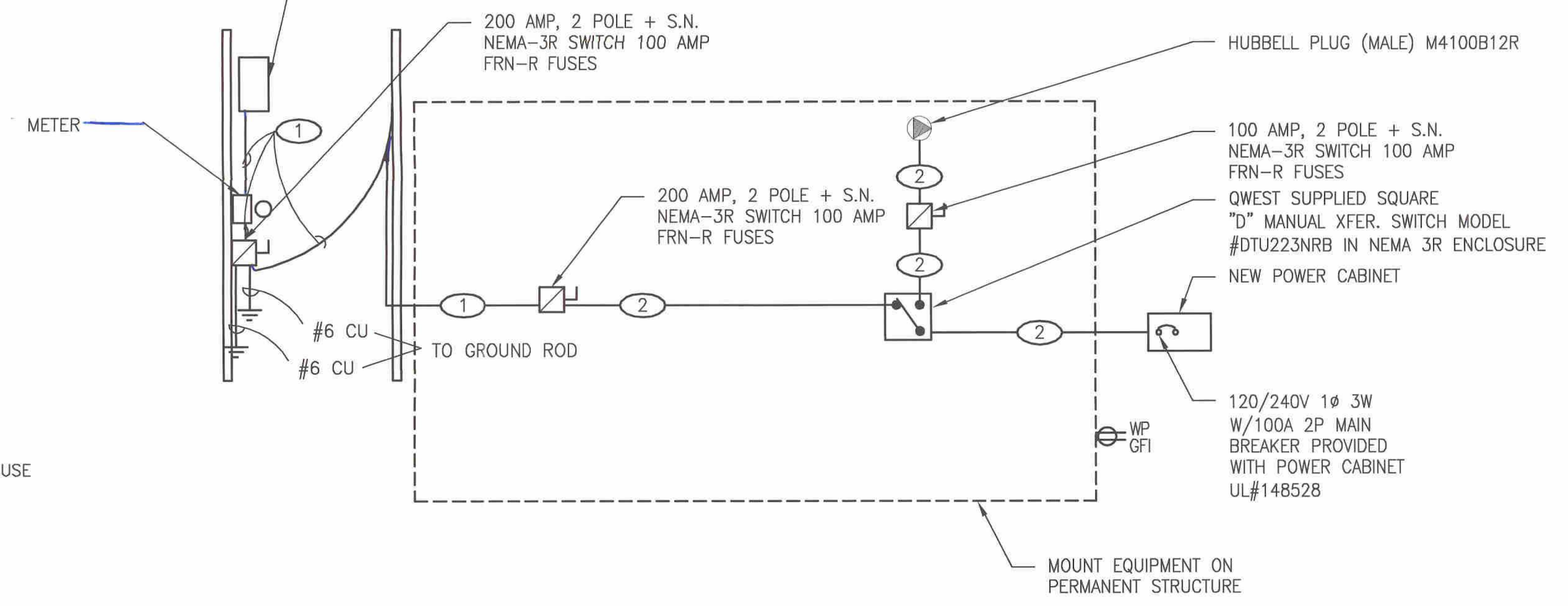

TYPICAL SECTOR
RX/TX
RX
ANTENNA
7-16 DIN FEMALE CONNECTOR ON ANTENNA
7-16 DIN MALE CABLE CONNECTOR
BUTYL RUBBER WEATHERPROOF KIT
FOR COLOR CODE.
COAX JUMPER
7-16 DIN MALE CABLE CONNECTOR
7-16 DIN FEMALE CABLE CONNECTOR
BUTYL RUBBER WEATHERPROOF KIT
ANDREW GROUND KIT
GROUND BAR (TOP OF MONOPOLE)
GROUND BAR (BOTTOM OF MONOPOLE)
ANDREW HELIAX CABLE TYPE (SEE COAXIAL CABLE SCHEDULE)
GROUND BAR (AT ANTENNA CABLE COVER ASSEMBLY)
FOR COLOR CODE.
7-16 DIN FEMALE CABLE CONNECTOR
7-16 DIN MALE CABLE CONNECTOR
PRIMARY RADIO CABINET INTERFACE PLATE.
ANDREW SUPER FLEX CABLE
"N" TYPE MALE COAX CONNECTOR
"N" TYPE FEMALE INTERFACE PLATE CONNECTOR
GPS ANTENNA
LUCENT GPS ANTENNA KIT #407517689
"N" TYPE FEMALE CABLE CONNECTOR
"N" TYPE MALE CABLE CONNECTOR
GROUND KIT IF RUN OVER 20 FEET
ANDREW HELIAX CABLE TYPE
GROUND KIT
ANTENNA CABLE COVER ASSEMBLY (NOT AT INTERIOR APPLICATIONS)
GROMMET INTO DOGHOUSE
"N" TYPE MALE CONNECTOR
"N" TYPE FEMALE INTERFACE PLATE CONNECTOR

### 720 ANTENNA COAX CABLE GROUNDING SCHEMATIC

SCALE : N.T.S.

## FEEDER SCHEDULE

| | |
|---|---|
| ① | (3#4/0 THWN CU) 2 1/2" C-NEW |
| ② | (3#1 THWN CU, 1#6 GR) 1 1/2" C-NEW |


EXISTING 50 KVA, 120/240 V 1ø POUDRE VALLEY IREA POLE MOUNTED TRANSFORMER
METER
200 AMP, 2 POLE + S.N. NEMA-3R SWITCH 100 AMP FRN-R FUSES
1
#6 CU
#6 CU
TO GROUND ROD
1
200 AMP, 2 POLE + S.N. NEMA-3R SWITCH 100 AMP FRN-R FUSES
2
HUBBELL PLUG (MALE) M4100B12R
2
100 AMP, 2 POLE + S.N. NEMA-3R SWITCH 100 AMP FRN-R FUSES
2
QWEST SUPPLIED SQUARE "D" MANUAL XFER. SWITCH MODEL #DTU223NRB IN NEMA 3R ENCLOSURE
NEW POWER CABINET
2
120/240V 1ø 3W W/100A 2P MAIN BREAKER PROVIDED WITH POWER CABINET UL#148528
WP GFI
MOUNT EQUIPMENT ON PERMANENT STRUCTURE

## ONE-LINE DIAGRAM

NO SCALE

**NOTE:**

1. CONTRACTOR TO FIELD VERIFY ALL DIMENSIONS, CABLE/CONDUIT LENGTHS AND ALL EXISTING CONDITIONS PRIOR TO CONSTRUCTION.
2. HEAVY LINE WEIGHT INDICATES NEW EQUIPMENT, ALL OTHER IS EXISTING.

| NO. | DATE | REVISION | BY | CHKD | APVLS |
|---|---|---|---|---|---|
| A | 10 12 01 | ISSUED FOR QWEST REVIEW | IR | RLS | |
| 0 | 10 23 01 | ISSUED FOR CONSTRUCTION/PERMIT | IR | RLS | |
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COLORADO REGISTERED PROFESSIONAL ENGINEER
WALTER RABINOVICH
29145
11/14/01

Qwest® Wireless™

SITE INFORMATION
SOUTHDOWN CEMENT PLANT
5134 UTE HIGHWAY
LYONS, COLORADO 80540-0529
BLD-021A

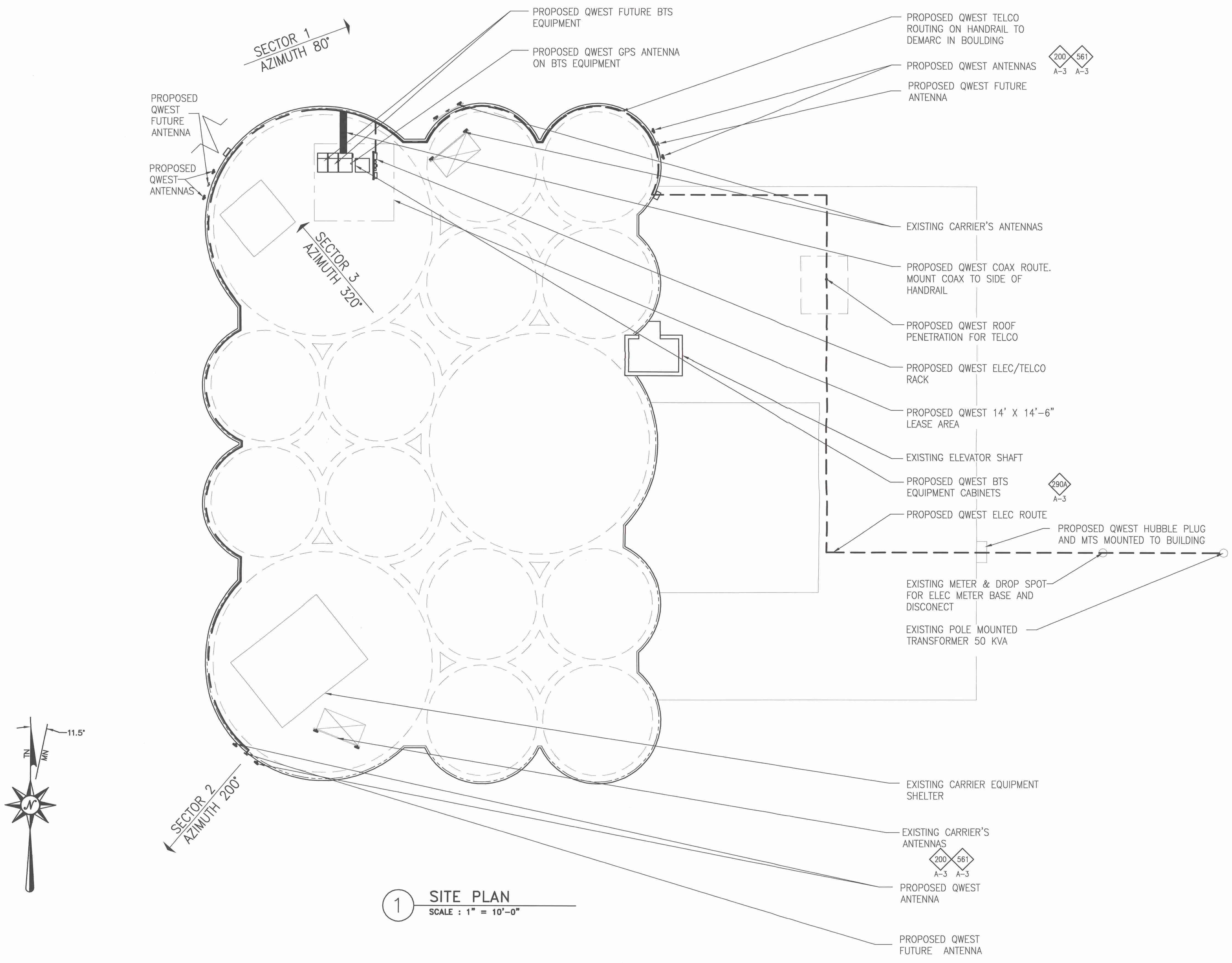
DESIGN TYPE
ROOFTOP EQUIPMENT
AND
ANTENNAS

SHEET TITLE
ONE-LINE DIAGRAM
CABLE COAX SCHEMATIC
AND SCHEDULES

SHEET NUMBER
E-1

REV.
0

DWG NAME: E1
DATE: 10-23-01


PROPOSED QWEST FUTURE BTS EQUIPMENT
PROPOSED QWEST GPS ANTENNA ON BTS EQUIPMENT
PROPOSED QWEST TELCO ROUTING ON HANDRAIL TO DEMARC IN BOULDING
PROPOSED QWEST ANTENNAS
200 A-3
561 A-3
PROPOSED QWEST FUTURE ANTENNA
SECTOR 1 AZIMUTH 80°
PROPOSED QWEST FUTURE ANTENNA
PROPOSED QWEST ANTENNAS
SECTOR 3 AZIMUTH 320°
EXISTING CARRIER'S ANTENNAS
PROPOSED QWEST COAX ROUTE. MOUNT COAX TO SIDE OF HANDRAIL
PROPOSED QWEST ROOF PENETRATION FOR TELCO
PROPOSED QWEST ELEC/TELCO RACK
PROPOSED QWEST 14' X 14'-6" LEASE AREA
EXISTING ELEVATOR SHAFT
PROPOSED QWEST BTS EQUIPMENT CABINETS
290A A-3
PROPOSED QWEST ELEC ROUTE
PROPOSED QWEST HUBBLE PLUG AND MTS MOUNTED TO BUILDING
EXISTING METER & DROP SPOT FOR ELEC METER BASE AND DISCONECT
EXISTING POLE MOUNTED TRANSFORMER 50 KVA
TN
MN
11.5°
N
SECTOR 2 AZIMUTH 200°
EXISTING CARRIER EQUIPMENT SHELTER
EXISTING CARRIER'S ANTENNAS
200 A-3
561 A-3
PROPOSED QWEST ANTENNA
PROPOSED QWEST FUTURE ANTENNA

1 SITE PLAN
SCALE : 1" = 10'-0"

| NO. | DATE | REVISION | BY | CHKD | APVLS |
|---|---|---|---|---|---|
| A | 10<br>12<br>01 | ISSUED FOR QWEST REVIEW | IR | RLS | |
| 0 | 10<br>23<br>01 | ISSUED FOR CONSTRUCTION/PERMIT | IR | RLS | |
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WALTER RABINOVICH
29145
11/14/01

Qwest® Wireless™

SITE INFORMATION

SOUTHDOWN CEMENT PLANT
5134 UTE HIGHWAY
LYONS, COLORADO 80540-0529
BLD-021A

DESIGN TYPE

ROOFTOP EQUIPMENT
AND
ANTENNAS

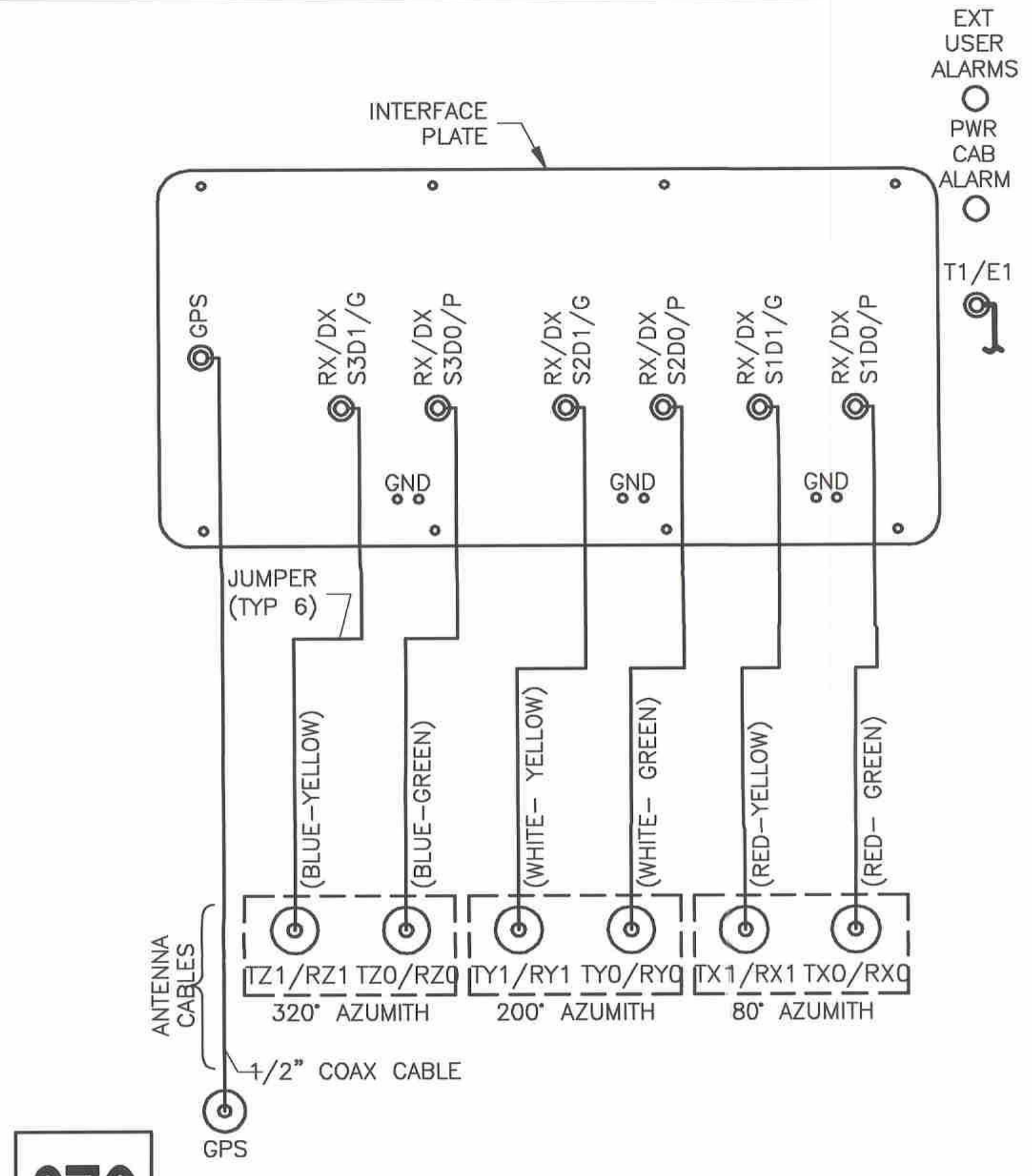
SHEET TITLE

ELECTRICAL SITE PLAN

SHEET NUMBER
E-2

REV.
0

DWG NAME: E-2
DATE: 10-23-01


EXT USER ALARMS
PWR CAB ALARM
INTERFACE PLATE
T1/E1
GPS
RX/DX S3D1/G
RX/DX S3D0/P
RX/DX S2D1/G
RX/DX S2D0/P
RX/DX S1D1/G
RX/DX S1D0/P
GND
GND
GND
JUMPER (TYP 6)
(BLUE–YELLOW)
(BLUE–GREEN)
(WHITE– YELLOW)
(WHITE– GREEN)
(RED–YELLOW)
(RED– GREEN)
ANTENNA CABLES
TZ1/RZ1 TZ0/RZ0
320° AZUMITH
TY1/RY1 TY0/RY0
200° AZUMITH
TX1/RX1 TX0/RX0
80° AZUMITH
1/2" COAX CABLE
GPS

## 676 COAX INTERFACE PLATE AT PRIMARY CABINET - E-CELL LAYOUT

SCALE: N.T.S.

COLOR CODE THE COAXIAL CABLES WITH COLORED ELEC. TAPE AS DESCRIBED BELOW. THE ACTUAL COLOR CODING WILL TAKE PLACE AFTER THE CABLE IS LAID OUT AND THE CONNECTORS ARE INSTALLED, SO THE COLOR CODING WILL OCCUR DURING THE RETURN LOSS MEASUREMENT ON THE COAX.

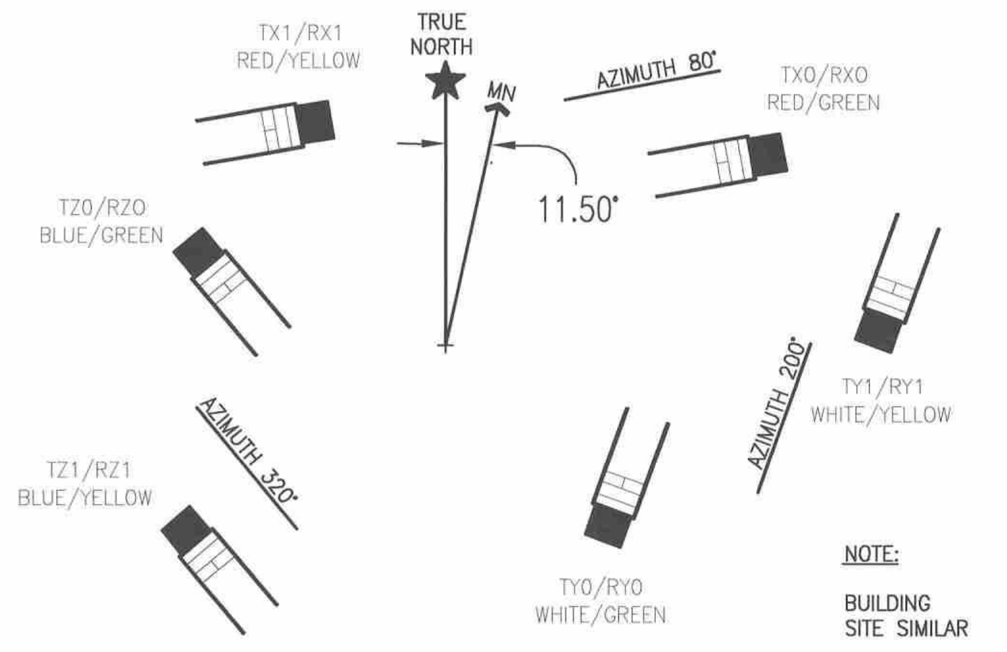
THE COLOR CODING IS AS FOLLOWS:

TO IDENTIFY THE SECTOR:
X = RED 80 DEGREES OR TRUE NORTH
Y = WHITE 200 DEGREES OR E.S.E.
Z = BLUE 320 DEGREES OR W.S.W.

TO IDENTIFY THE PORT:
0 = GREEN
1 = YELLOW

SO EACH COAX WILL HAVE TWO STRIPES ON EACH END OF THE CABLES. THESE STRIPES SHOULD BE APPROXIMATELY TWO INCHES LONG.

FOR THE END OF THE COAX THAT GOES TO THE ANTENNA, ATTACH THE STRIPES AS NEAR TO THE ANTENNA AS POSSIBLE, ENSURING THAT THEY WON'T GET COVERED UP BY THE WEATHER–PROOFING AND CAN BE SEEN FROM THE GROUND.

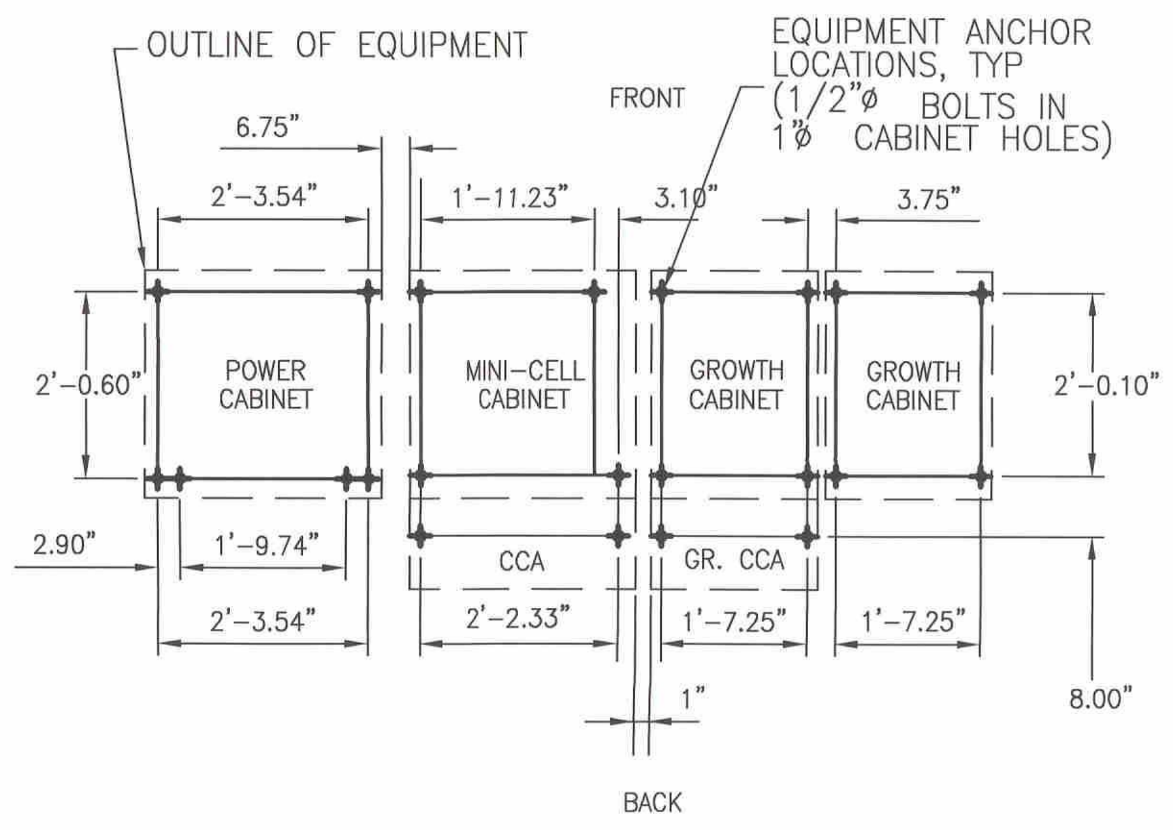

TX1/RX1 RED/YELLOW
TRUE NORTH
MN
AZIMUTH 80°
TX0/RX0 RED/GREEN
11.50°
TZ0/RZ0 BLUE/GREEN
AZIMUTH 320°
TZ1/RZ1 BLUE/YELLOW
AZIMUTH 200°
TY1/RY1 WHITE/YELLOW
TY0/RY0 WHITE/GREEN

NOTE:

BUILDING SITE SIMILAR

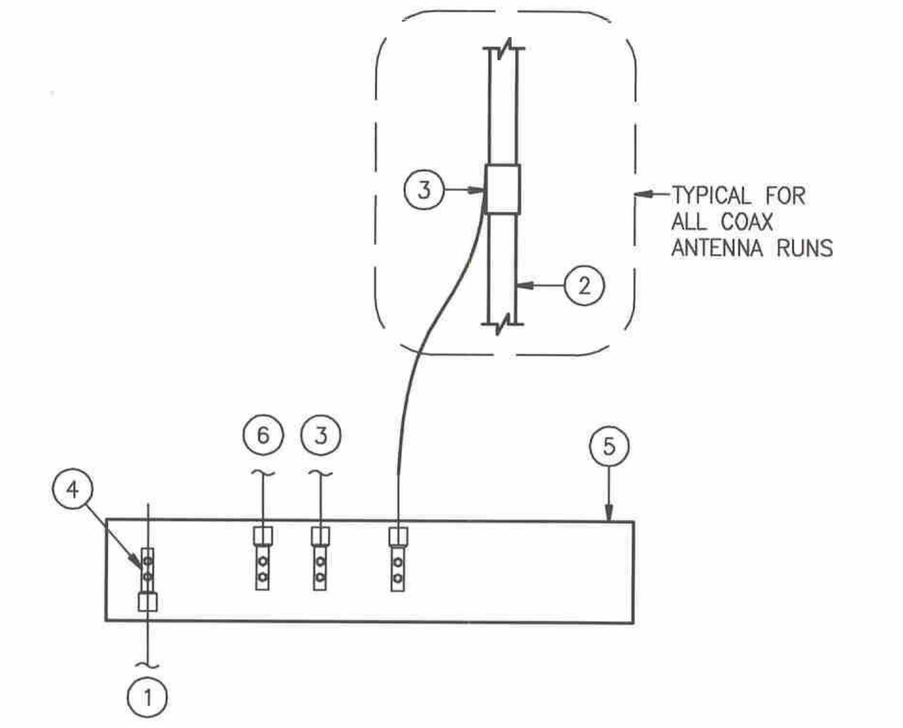
## 680 COAX CABLE COLOR CODING

SCALE: N.T.S.


OUTLINE OF EQUIPMENT
EQUIPMENT ANCHOR LOCATIONS, TYP (1/2"ø BOLTS IN 1"ø CABINET HOLES)
FRONT
6.75"
2'–3.54"
1'–11.23"
3.10"
3.75"
2'–0.60"
POWER CABINET
MINI–CELL CABINET
GROWTH CABINET
GROWTH CABINET
2'–0.10"
2.90"
1'–9.74"
CCA
GR. CCA
2'–3.54"
2'–2.33"
1'–7.25"
1'–7.25"
1"
8.00"
BACK

## 910 EQUIPMENT ANCHOR PATTERN

SCALE: N.T.S.


③
②
TYPICAL FOR ALL COAX ANTENNA RUNS
④
⑥
③
⑤
①

**KEY NOTES:**

1) #2 AWG BCW TO GROUND BAR AT EQUIPMENT
2) ANTENNA COAX CABLE
3) STANDARD ANDREWS COAX GROUND KIT, TYPICAL AT EACH ANTENNA DO NOT INSTALL ON BEND
4) 2 HOLE LUG TYPE CONNECTOR – TYPICAL
5) GROUND BAR MOUNTED AT EQUIPMENT SEE DETAIL 710
6) FUTURE ANTENNA COAX GROUND

## 750 GROUND BAR AT ANTENNAS

SCALE: N.T.S.

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|---|---|---|---|---|---|
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| 0 | 10<br>23<br>01 | ISSUED FOR CONSTRUCTION/PERMIT | IR | RLS | |
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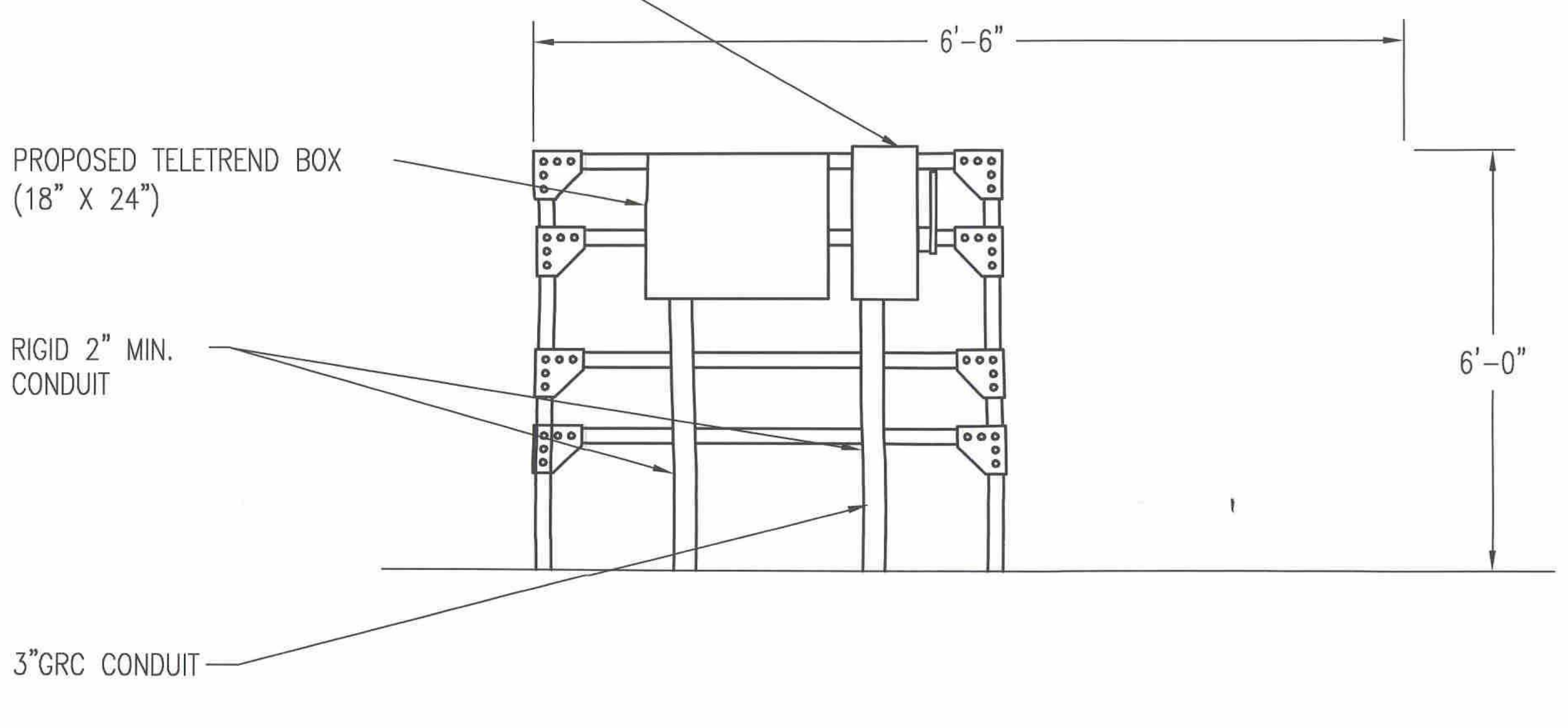
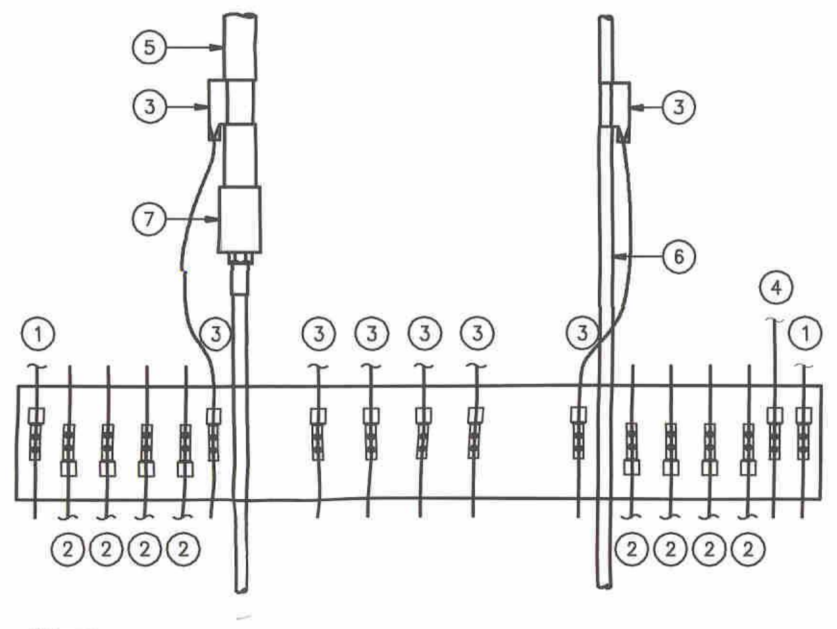

PROPOSED DISCONNECT (6 3/8"X15"X6")
6'–6"
PROPOSED TELETREND BOX (18" X 24")
RIGID 2" MIN. CONDUIT
6'–0"
3"GRC CONDUIT

NOTES:
1. SIZE CONDUIT PER ONE–LINE DIAGRAM ON SHEET E–1.
2. ATTACH CONDUIT TO FRAME PER NATIONAL ELECTRICAL CODE.
3. TELTREND BOX 2 EAC H 6 X 22 BSW TO PEDESTAL
4. ADD " DOUBLE UNISTRUT ON VERTICAL & HORIZONTAL MEMBERS" PART NO.P1001
5. EITHER SINGLE 90° GUSSETS PART NO.P1873 OR 2 "T" PART NO.P1380A BACK TO BACK ON EITHER SITE

## 951N ELEC/TELCO RACK

SCALE: N.T.S.


⑤
③
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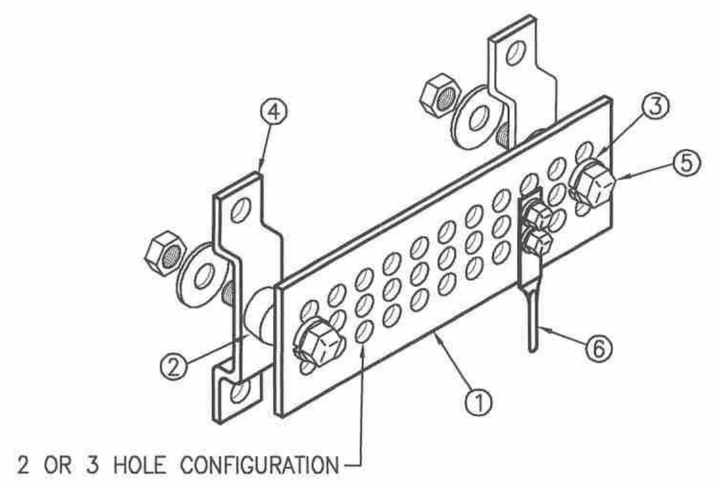
②②②②
②②②②

**NOTES:**

1) #2 AWG BCW TINNED TO EXISTING LIGHTNING PROTECTION SYSTEM, (2) TOTAL
2) #2 AWG BCW TO EQUIPMENT CABINETS, AS FOLLOWS, 8 TOTAL SPACES
    - 2 – POWER CABINET
    - 2 – RADIO CABINET
    - 2 – GROWTH CABINET (FUTURE)
    - 2 – GROWTH CABINET (FUTURE)
3) STANDARD ANDREWS COAX GROUND KIT, TYPICAL AT COAX DO NOT INSTALL ON BEND
4) #2 AWG BCW TO GROUND CONDUCTOR AT POLE
5) ANTENNA COAX, 4 TOTAL, 1 SHOWN
6) GPS COAX
7) COAX TO JUMPER CONNECTOR

## 752 GROUND BAR AT EQUIPMENT

SCALE: N.T.S.


④
③
⑤
②
⑥
①
2 OR 3 HOLE CONFIGURATION

NOTE: TWO HOLE LUG CONNECTOR BOLTED TO EACH SIDE OF GROUND BAR ACCEPTABLE TO REDUCE GROUND BAR SIZE.

KEY NOTES:

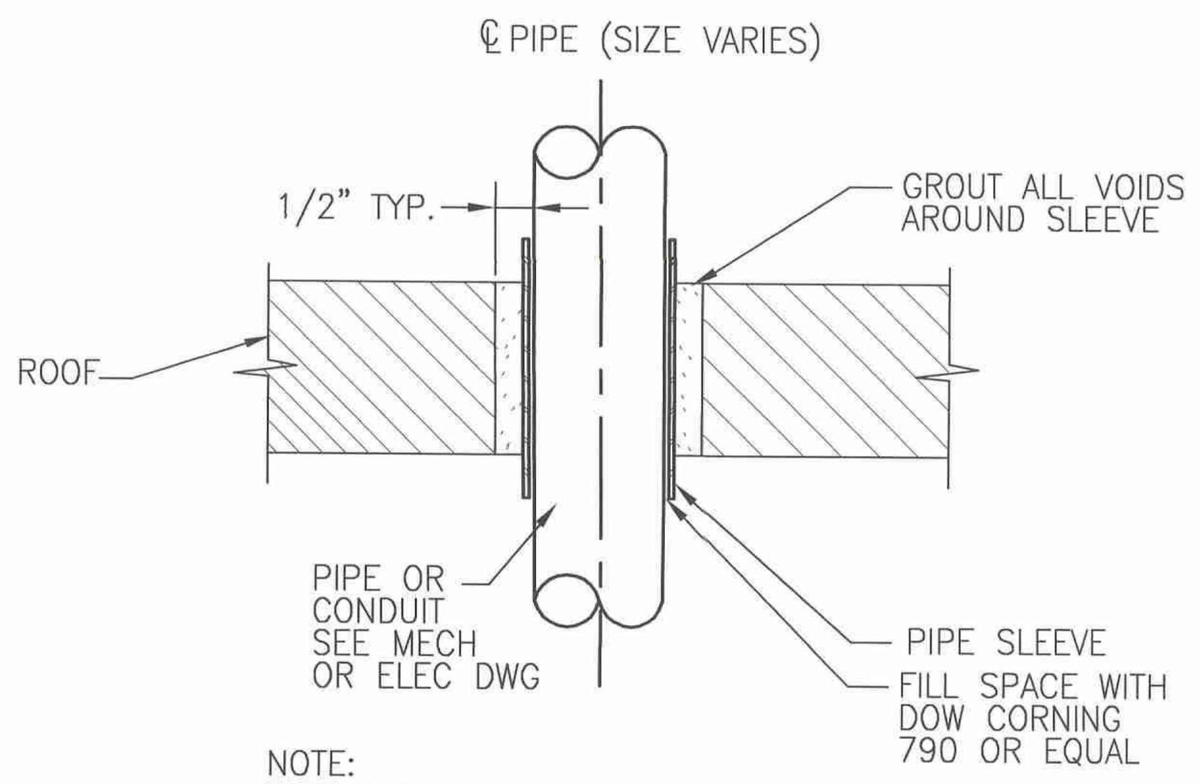
1) COPPER GROUND BAR
1/4" x 4" x LENGTH AS REQUIRED, HOLE CENTERS TO MATCH NEMA DOUBLE LUG CONFIGURATION.
2) SPACERS AS REQUIRED – NO INSULATORS
3) 5/8"ø LOCK WASHERS
4) WALL MOUNTING BRACKET
5) 5/8"ø H.H.C.S. BOLTS
6) TWO HOLE LUG TYPE CONNECTOR TYPICAL.

## 710 STANDARD GROUNDING BAR

SCALE: N.T.S.

WALTER RABINOVICH 29145

11/19/01


℄ PIPE (SIZE VARIES)
1/2" TYP.
GROUT ALL VOIDS AROUND SLEEVE
ROOF
PIPE OR CONDUIT SEE MECH OR ELEC DWG
PIPE SLEEVE
FILL SPACE WITH DOW CORNING 790 OR EQUAL

NOTE:
PIPE ANCHORAGE IS REQUIRED NEARBY TO PREVENT PIPE MOVEMENT THRU PENETRATION. CABLES WITHIN CONDUIT PENETRATION SHALL BE APPROPRIATELY SEALED AFTER INSTALLATION.

## 632A ROOF PENETRATION

SCALE: N.T.S.

Qwest® Wireless™

SITE INFORMATION

SOUTHDOWN CEMENT PLANT
5134 UTE HIGHWAY
LYONS, COLORADO 80540–0529
BLD–021A

DESIGN TYPE

ROOFTOP EQUIPMENT
AND
ANTENNAS

SHEET TITLE

DETAILS

SHEET NUMBER: E–3

REV.: 0

DWG NAME: E–3

DATE: 10–23–01

FINAL

RECEIVED
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BOULDER COUNTY
BUILDING DIV.

I-2179

U-2

COUNTY COPY

Qwest
5734 Ute Hwy

PLAN REVIEW
& PERMIT SECTION ☐
ACCEPTED SUBJECT ☐
TO CORRECTION NOTED ☐
REFUSED
FINAL ACCEPTANCE SUBJECT ☐
TO FIELD INSPECTION
BY Larry DATE: 12/20/1
PERMIT NO.